

Fitness App: Illustrated AI Reports

Generated 2026-10-09 UTC from the current local code and trained artifacts.

Seven module reports explain inputs, processing, training, examples, measured evidence, and limits. No patient or user records are included. Research-based rules and synthetic labels are not independent human-expert validation.

TrainCore / Workout Recommendations

Local Random Forest regressor

NutriCore / Meals And Fridge Swaps

Local Random Forest classifier

PredictionAI / Weight Trend

Local ordinary least-squares linear regression

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GoalAI / Suggested Goal Weight

Explicit body-composition heuristics

AI Coach / Conversation

External Gemini model through a PHP API wrapper

TrainCore / Workout Recommendations

Local Random Forest regressor



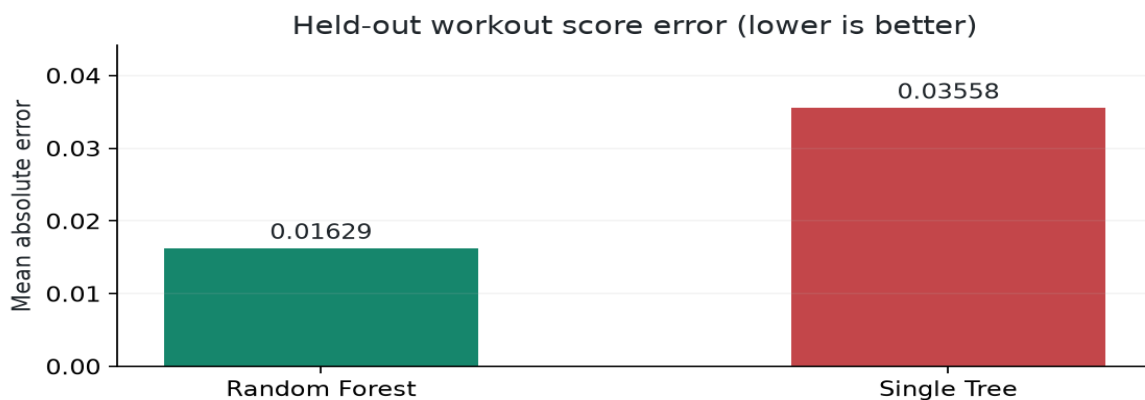
How It Works

Encode goal, location, intensity, exercise category, movement family, and numeric fit features. Each trained tree predicts an exercise suitability score; average the 64 predictions. Hard restrictions and plan-building rules remain separate from the learned model.

Training And Evidence

Rule-derived profile/exercise examples: 25,408 training and 6,118 held out. Grouped profiles stay together. Compare the forest with a single-tree baseline. No feature scaling is required.

Held-out score MAE: 0.01629; single-tree baseline: 0.03558. The training pipeline generated and checked 3,960 profile plans. These are rule-agreement and structural checks, not clinical validation.



Example

Illustrative: a home-training user with knee pain gets location- and injury-filtered candidates before scoring. A high model score cannot override those restrictions. The selected exercises are arranged into the requested weekly schedule.

Limits

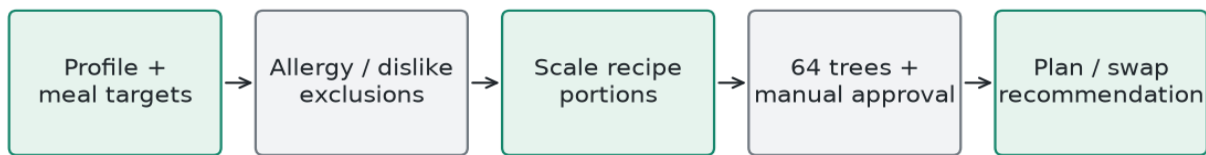
Rule-derived labels are not observed injury or fitness outcomes. The forest is easier to describe than a neural network, but 64 trees are less transparent than one tree. Check equipment availability and obtain appropriate professional advice for injuries.

Source

`ml/workout_ai/traincore_model.py`; `ml/workout_ai/traincore_trained_model.json`

NutriCore / Meals And Fridge Swaps

Local Random Forest classifier



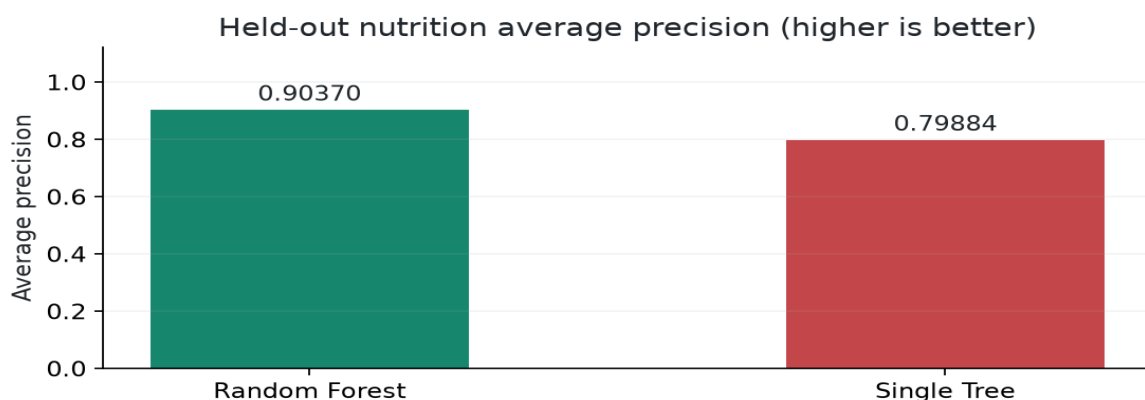
How It Works

Preprocess complete USDA nutrient records into 72 recipes. Scale portions within 0.5x-2x, keeping ingredient grams and nutrients consistent. Trees compare macro, goal, fridge, and preference fit. Only manually approved options can be shown. Small macro bonuses, feedback, and variety adjustments influence final ranking; duplicate aliases are suppressed.

Training And Evidence

39,481 rule-derived examples over 600 scenarios. Split scenarios 60/20/20 for fitting, development, and test; select settings on development only, then refit on fitting plus development. Tied rubric scores receive consistent labels, and identical recipes share training weight.

Held-out average precision: 0.9037; baseline: 0.79884. Top-three rule-label hit rate: 98.84% over 431 eligible test slots. Human training ratings currently available: 0.



Example

Illustrative: with a 2,200-calorie daily target and four meals, lunch receives 30%, or 660 calories. A 550-calorie recipe scales by 1.2. A base 40 g protein becomes 48 g. Milk-containing options are excluded for a recorded dairy allergy. An unapproved recipe is excluded even with a high score.

Limits

Scores are relative suitability, not health probabilities. Existing recipe approval is not expert validation of every personalized portion. Gram instructions state raw/dry, cooked, drained, or as-sold weight. Ingredient records do not guarantee absence of allergen cross-contact. Macro totals alone do not establish fibre, sodium, or micronutrient adequacy.

Source

ml/nutrition_ai/nutricore_model.py; train_model.py; MODEL_REPORT.md; REVIEW_GUIDE.md

PredictionAI / Weight Trend

Local ordinary least-squares linear regression



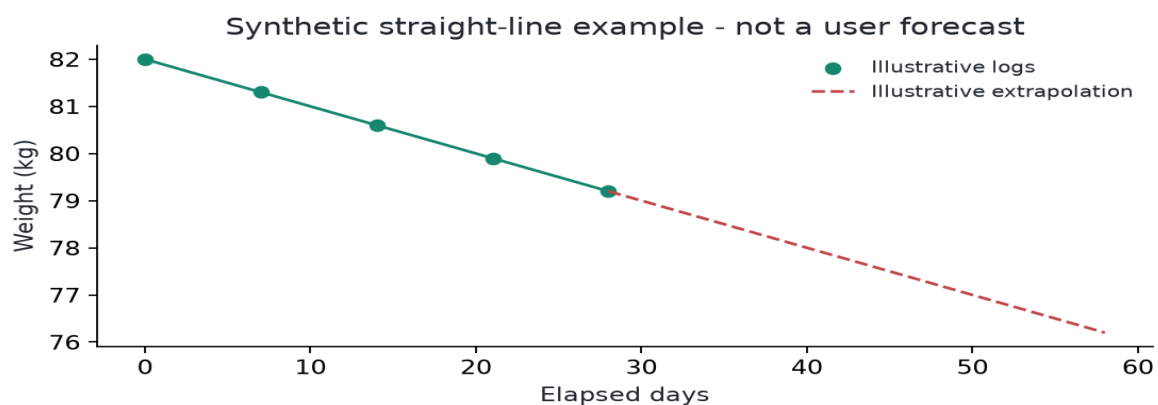
How It Works

Fit weight = intercept + slope x elapsed days. The slope describes average weight change per day. Fit the final trend to all available records and extrapolate 30 days beyond the latest observation. Refit for each user when requested; no shared model artifact needs offline training.

Training And Evidence

With at least four records and two distinct dates in the earlier training subset, hold out the latest 25% and predict those observations using only earlier data. Validation MAE is unavailable when there is insufficient data. Legacy training_accuracy is R-squared percentage, not classification accuracy.

The local validation script passes 10 cases covering exact lines, chronological error, ordering, insufficient data, and duplicate dates. A perfect score on a synthetic straight line is not real-world forecast accuracy.



Example

Illustrative only: logs following weight = 82 - 0.1 x day have a slope of -0.1 kg/day. If the latest day is 28, a 30-day extension reaches day 58 and predicts 76.2 kg. This is mathematical extrapolation, not a promised outcome.

Limits

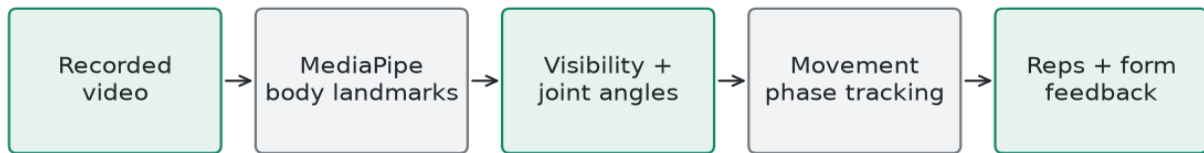
Weight change is rarely linear indefinitely. Water fluctuations, sparse logs, changing behaviour, and medical factors can invalidate the forecast. It is a trend estimate, not a treatment or target-weight prescription.

Source

backend/services/PredictionAI.php; scripts/test-prediction-ai.php

PoseForm / Exercise Video

Pretrained MediaPipe plus explicit movement rules



How It Works

Read video frames and locate body joints with pretrained MediaPipe. Compute joint angles and select the more visible side. Exercise-specific start/end phases, range checks, and visibility thresholds determine repetitions and form feedback. This module still uses a neural pose detector; it was not replaced or retrained.

Training And Evidence

No local neural-network training. The supplied MediaPipe model is already trained. Local pose templates and movement rules are validated using synthetic pose sequences and exercise aliases.

17 template/form tests, two JavaScript repetition/noise checks, and three Python analyzer suites pass. An uploaded-video smoke test processed 62 landmark frames across 24 exercise modes. Running all modes on one clip checks robustness, not accuracy for 24 different exercises.

Example

Illustrative: a squat must complete the configured standing-to-lowered-to-standing sequence. Small jitter and incomplete movement should not count as a full repetition. If body landmarks are unclear, the analyzer requests a clearer recording rather than treating missing joints as valid movement.

Limits

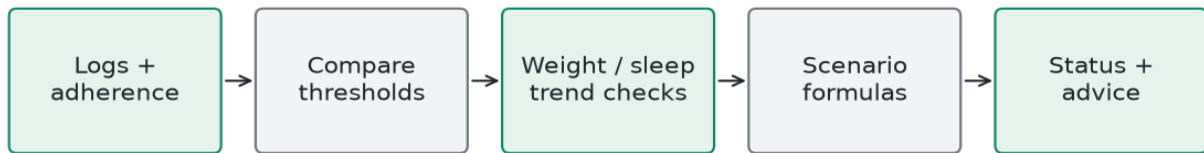
Camera angle, occlusion, lighting, frame skipping, exercise selection, and movement variation affect results. Labeled exercise videos are needed to measure rep-count and form accuracy independently.

Source

ml/exercise_ai/video_pose_analyzer.py; poseform_pose_templates.json; scripts/validate-exercise-ai.js

ProgressAI / Progress And Scenarios

Explicit PHP rules and heuristic simulation



How It Works

Compare workout and nutrition adherence with explicit thresholds, inspect recent weight change, and generate predefined status and advice. Scenario simulation combines calorie delta, steps, sleep, and workout frequency using project formulas. There is no learned model or training file.

Training And Evidence

Not applicable. Rules are edited and tested as ordinary code; weights are not learned from examples.

This report describes the current source. The recommendation test suites do not independently validate ProgressAI's simulated health outcomes. No predictive accuracy is claimed.

Example

With 50% workout adherence and 80% nutrition adherence, the audit reports Needs Attention because at least one adherence value is below 60%. Advice emphasizes consistency rather than adding extra training volume.

Limits

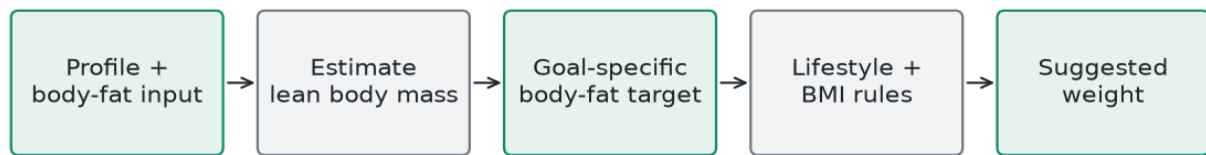
Simulation constants and thresholds are project heuristics. They are not individualized physiology or clinical evidence, and status labels should not be treated as diagnoses.

Source

backend/services/ProgressAI.php

GoalAI / Suggested Goal Weight

Explicit body-composition heuristics



How It Works

Estimate lean body mass from weight and supplied or heuristically estimated body fat. Compute a suggested weight for a fixed goal-specific body-fat percentage, apply small lifestyle adjustments, then apply BMI-related safeguards. No neural network or Random Forest is used.

Training And Evidence

Not applicable. The body-fat estimates, targets, and adjustments are hand-written formulas, not learned parameters.

No independent validation of goal-weight accuracy is available. Reported values describe the current code, not professional recommendations.

Example

Illustrative intermediate calculation: 80 kg at 25% body fat implies 60 kg lean mass. A hypothetical 20% target gives $60 / 0.8 = 75$ kg before adjustments. That target is an illustration, not an actual recommendation for a user.

Limits

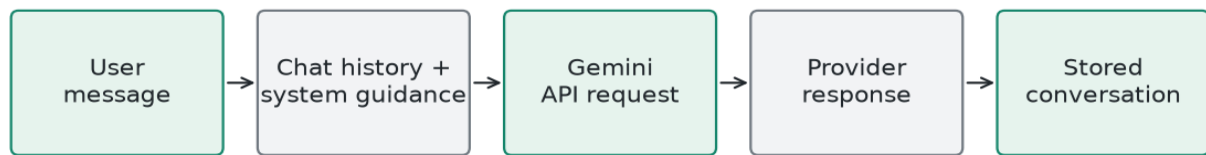
Waist-only estimates and default body-fat assumptions can be inaccurate. Fixed body-fat targets and BMI rules require professional review before presenting them as health-optimal goals. The app's suggested weight is not a clinical prescription.

Source

backend/services/GoalAI.php

AI Coach / Conversation

External Gemini model through a PHP API wrapper



How It Works

Send conversation history and coaching instructions to Gemini using a server-side environment key. Display the provider response and store chat history. The chat instructions reserve workouts, meals, and video analysis for the dedicated local modules.

Training And Evidence

No local model training or fine-tuning. GEMINI_MODEL chooses the provider model. A model-name alias may change over time; it is not a reproducible locally trained artifact.

Requires configured credentials and network access. This report does not make a live Gemini call or claim an evaluated coaching accuracy. Credentials are excluded from Git and the report.

Example

A question about staying consistent is sent as conversational context. Full workout generation remains TrainCore's responsibility; NutriCore owns meal plans, and PoseForm owns recorded-video analysis.

Limits

Provider responses can be incorrect or inappropriate. Conversation content is transmitted to an external service. Use privacy-aware inputs and do not treat chat as medical diagnosis.

Source

backend/controllers/AIChatController.php; backend/services/GeminiService.php